SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY

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An IOT-Based Smart Doorbell System Using Camera and Ultrasonic Sensor for Real-Time Visitor Detection and Context-Aware Monitoring

Academic Year : 2025-2026 **Year & Sem : IV-II**

ABSTRACT

The Smart Doorbell System improves home security using Python and Arduino. An ultrasonic sensor detects visitors, and a camera sends images via Telegram for user verification. The user can unlock the door remotely, triggering a motor to open it for 20 seconds before closing. Safety features like a GSM module, buzzer, and LED provide alerts, all powered by a 12V system.

DIS ADVANTAGES OF EXISTING SYSTEM

- Traditional Wired Doorbell Systems
- Wireless Doorbell Systems
- Manual Button-Based Alert Systems
- Motion Sensor-Based Doorbells
- Camera-Based Smart Doorbell Systems

EXISTING SYSTEM METHODS

- Requires manual operation
- Possibility of button failure or damage
- No automatic visitor detection
- Limited security and monitoring features

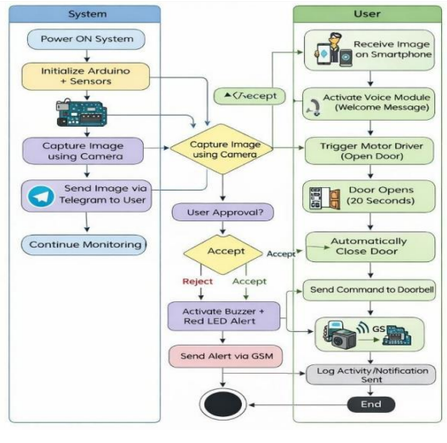
PROPOSED SYSTEM

The system uses an ultrasonic sensor and Arduino to detect visitors near the door in real time. When someone is within range, it automatically triggers a buzzer alert. It offers a simple, low-cost solution for home security, with potential for future IoT integration. Additionally, it operates continuously to ensure reliable monitoring.

ADVANTAGES OF PROPOSED SYSTEM

- Real-time alert generation using buzzer
- Low-cost and simple implementation using Arduino
- Improved home security and convenience

FLOW CHART



MODULES

- Ultrasonic Sensor Detection Module
- Distance Measurement Module
- Microcontroller Processing Module
- Alert Generation Module (Buzzer)

MODULES DESCRIPTION

The ultrasonic sensor detection module measures the distance between the sensor and nearby objects to identify the presence of a person near the door.

The distance measurement module processes the echo signals from the ultrasonic sensor to calculate the exact distance of the detected object.

OUTPUTS

CONCLUSION

In summary, the Smart Doorbell System using an ultrasonic sensor provides an efficient solution for automatic visitor detection and alert generation. The system continuously monitors the area near the door, measures distance using the ultrasonic sensor, and detects the presence of a person in real time. By integrating the sensor with a microcontroller, the system triggers a buzzer alert without requiring manual interaction.

BATCH NO: B – 15

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